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A hybrid approach to time series forecasting: Integrating ARIMA and prophet for improved accuracy

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ABSTRACT

This research investigates the critical role of time series forecasting within the burgeoning field of edge computing. With the increasing volume of data generated by IoT devices, performing analytics at the edge offers significant advantages in terms of latency, bandwidth consumption, and privacy. This paper explores various forecasting models and algorithms applied to time series data in edge environments, focusing on both univariate and multivariate approaches. We delve into the specifics of univariate time series forecasting using the Prophet model, highlighting its strengths in handling seasonality and trend. For multivariate time series, we examine the Vector Autoregressive Integrated Moving Average (VARIMA) model, a powerful tool for capturing interdependencies between multiple time series. Recognizing the limitations of individual models, we also investigate a hybrid time series forecasting approach, combining the strengths of ARIMA (Autoregressive Integrated Moving Average) and Prophet. This hybrid model enhances forecast accuracy by leveraging ARIMA's ability to capture linear dependencies and short-term fluctuations, while simultaneously utilizing Prophet's effectiveness in modelling non-linear trends, seasonality, and holiday effects. By integrating these complementary strengths, our approach provides a more robust and adaptive framework that consistently outperforms individual models, especially in complex time series exhibiting both stationary and non-stationary components. A key aspect of this work involves evaluating the performance of these models using RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error. Experimental results are presented to showcase the effectiveness and comparative performance of the discussed forecasting techniques across diverse simulated and real-world datasets relevant to edge computing applications.

1. Introduction

1.1. The role of edge computing in forecasting

Edge computing plays a pivotal role in enhancing forecasting algorithms and models by bringing data processing closer to the source of data generation [1]. In traditional cloud-based systems, data is often sent to centralized servers for analysis, which can introduce latency and reduce the timeliness of predictions. With edge computing, forecasting models can be deployed directly on local devices, such as sensors, gateways, or edge servers, enabling real-time analysis and predictions. This proximity to the data source allows for faster decision-making, reduced bandwidth usage, and improved privacy, as sensitive data doesn't need to be transmitted over long distances [2]. Additionally, edge computing supports scalability and resilience, as localized processing reduces dependency on central systems as visualised in Fig. 1. In time-sensitive applications like industrial automation, smart cities, or autonomous vehicles, edge computing enhances the effectiveness of forecasting algorithms by ensuring that predictions are timely, accurate, and actionable, even in environments with limited or intermittent connectivity to the cloud.

Time series forecasting algorithms are increasingly becoming pivotal in edge computing environments, where processing data closer to its source offers significant advantages [3]. Edge computing, which involves performing data processing on local devices rather than relying solely on centralized cloud servers, is evolving rapidly. Over the next decade, advancements in edge computing are expected to enhance its capabilities, allowing for more sophisticated and real-time data analysis. In this context, forecasting models play a crucial role in time series data by predicting future values based on historical trends. These models,

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including ARIMA, Prophet, and machine learning techniques, can be effectively deployed at the edge to provide timely and actionable insights [4]. As edge computing evolves, its integration with robust forecasting algorithms will enable more efficient and responsive decision-making, optimizing operations across various industries such as manufacturing, transportation, and smart cities.

1.2. Univariate and multivariate approaches in time series forecasting

Univariate and multivariate time series forecasting algorithms are essential tools in predicting future values based on past data, with applications across diverse fields such as finance, economics, healthcare, and engineering. Univariate time series forecasting involves models that predict future values using only a single variable, focusing on capturing trends, seasonality, and patterns within that one series. Common algorithms in this category include ARIMA, Simple Exponential Smoothing, and Prophet, which are effective in scenarios where the underlying patterns are consistent and stable. On the other hand, multivariate time series forecasting leverages multiple variables or features to make predictions, capturing the interdependencies and relationships between them. This approach is particularly useful when the time series is influenced by external factors or when multiple related series evolve together over time. Multivariate models, such as Vector Autoregression (VAR) and multivariate machine learning algorithms [5], provide a more comprehensive view, offering enhanced accuracy in complex scenarios where single-variable models might fall short. Both univariate and multivariate forecasting algorithms are critical in addressing the diverse challenges posed by time-dependent data, enabling more informed and accurate predictions.

1.3. Datasets and data analysis

Effective time series forecasting depends heavily on both the quality of the dataset and the thoroughness of the data analysis. To ensure accurate predictions, datasets should be comprehensive, containing a sufficient amount of historical data that captures the underlying patterns and trends. The frequency of data points must align with the forecasting needs, whether that's high-frequency data for short-term forecasts or lower frequency for long-term ones. Consistency in data intervals and completeness are crucial, with any missing values needing appropriate handling through imputation or estimation [6]. Additionally, the data must be relevant, with a focus on the target variable and any external factors that could impact the forecast. For many forecasting models, particularly those like ARIMA, the data should be stationary, meaning its statistical properties should remain constant over time, which might require transformations or differencing.

1.4. Time series forecasting at the edge

The proliferation of edge computing has enabled real-time data processing and decision-making closer to the data source, unlocking new possibilities for various applications. A critical aspect of leveraging edge data lies in the ability to forecast future trends and behaviors [7]. This research explores the diverse applications of time series forecasting within edge computing across multiple domains, as visually represented in Fig. 2. The figure highlights how edge computing, acting as the central

Fig. 1. Edge-Enabled IoT Infrastructure.

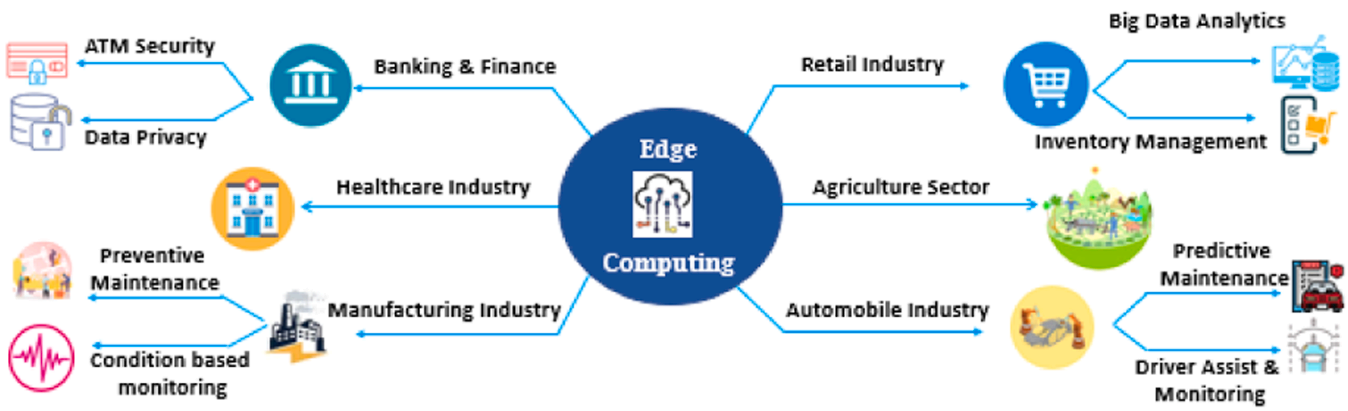

Fig. 2. Forecasting the Future at the Edge.

nervous system, connects various sectors and facilitates advanced analytics like time series forecasting to drive intelligent operations.

In banking and finance, edge computing strengthens ATM security by predicting potential breaches and ensures data privacy through proactive anonymization. Retailers optimize inventory management by forecasting demand and personalize customer experiences by predicting preferences [8]. Healthcare benefits from predictive maintenance of medical equipment and remote patient monitoring through condition-based forecasting. Manufacturing utilizes edge computing for predictive maintenance of machinery and real-time condition monitoring to ensure quality. The agriculture sector employs it for precision farming, optimizing resource utilization through weather and crop forecasting. Finally, in the automotive industry, edge computing enhances driver assist and monitoring systems by predicting driver behavior and enables predictive maintenance of vehicle components.

1.5. The significance of edge computing in real-time series analysis

Beyond its demonstrated accuracy and robustness, the proposed hybrid ARIMA-Prophet forecasting framework, particularly its suitability for deployment in edge computing environments, unlocks significant potential across a range of time-sensitive and data-intensive application areas. In smart cities, for instance, the model's ability to provide real-time traffic predictions at local intersections or transport hubs enables dynamic signal optimization and congestion management, significantly reducing latency compared to cloud-based systems. Similarly, within industrial IoT and manufacturing, this approach facilitates on-site predictive maintenance by forecasting equipment anomalies directly from sensor data, allowing for immediate intervention and minimizing costly downtime without continuous data offloading. For autonomous vehicles, where instant decision-making is paramount, edge-deployed hybrid models can forecast pedestrian movements or traffic flow changes with minimal delay, ensuring enhanced safety and navigation efficiency. Furthermore, in resource management, localized forecasting of energy demand or supply from distributed renewable sources can optimize load balancing and reduce transmission losses, benefiting from reduced bandwidth usage and improved system resilience at the edge. The integration of high-accuracy models with edge capabilities thus positions this framework as a crucial tool for transforming various industries that rely on timely, actionable insights derived from continuous data streams.

1.6. Data analysis

Data analysis plays a critical role in preparing the dataset for accurate forecasting. Exploratory Data Analysis (EDA) helps in understanding the basic characteristics of the data through visual inspection and statistical summaries. Identifying trends and seasonal patterns is essential, often involving decomposition of the time series into trend, seasonal, and residual components [9]. Correlation analysis, including autocorrelation and cross-correlation, aids in understanding dependencies and relationships between variables. Selecting and validating models involves evaluating various forecasting techniques and assessing their performance through metrics such as Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE), along with techniques like cross-validation. Feature engineering, including creating lagged features and incorporating external variables, enhances the model's ability to capture relevant information. Finally, error analysis and model refinement, including residual examination and parameter tuning, are crucial for improving the model's accuracy and robustness.

2. Comparison of forecating models and algorithms in timeseries data

2.1. Forecasting models and algorithms in time series data

Table 1 gives a brief comparison and summarizes several common time series forecasting models and algorithms, outlining their descriptions, typical use cases, and relevant model evaluation metrics. The models range from simpler approaches like Autoregressive (AR) and Moving Average (MA) models, suitable for specific data characteristics, to more complex models like ARMA and ARIMA, designed to handle a broader range of time series properties.

2.2. Characteristics of forecasting models and algorithms in time series data

A comparative overview of diverse time series forecasting models, detailing their key characteristics for informed selection, is provided in Table 2. The table considers data requirements, ranging from univariate to multivariate and the need for specific data transformations. Model complexity is assessed, considering computational demands and parameter tuning. Scalability addresses the model's ability to handle large datasets. Training time reflects the computational cost of model fitting. Interpretability highlights the ease of understanding the model's output and the influence of different factors. Long-term forecasting capabilities are compared, acknowledging the challenges of predicting far into the future. Assumptions regarding the error distribution and the model's robustness to outliers are also detailed. To improve forecast accuracy exogenous variables are compared to check whether the model can incorporate external factors. This comprehensive comparison facilitates informed model selection based on the specific characteristics of the time series data and the forecasting objectives.

Table 1
Comparison of existing time series forecasting models and its usecases.

Model/Algorithm	Description	Use Case	Model Evaluation Metrics	Reference
AR (AutoRegressive)	Exploits the relationship between a current data point and its preceding values.	Univariate time series, linear trend	MAE, MSE, RMSE, AIC, BIC	[10–12]
MA (Moving Average)	Captures the relationship between an observation and past forecast errors	Univariate time series with short-term noise	MAE, MSE, RMSE, AIC, BIC	[13,14]
ARMA (AutoRegressive Moving Average)	Integrates autoregressive and moving average processes into a single model	Stationary univariate time series	MAE, MSE, RMSE, AIC, BIC	[15,16]
ARIMA (AutoRegressive Integrated Moving Average)	Adds differencing to ARMA to make the series stationary.	Non-stationary univariate time series	MAE, MSE, RMSE, AIC, BIC, Dickey-Fuller test	[17–19]
SARIMA (Seasonal ARIMA)	Extends ARIMA to model seasonal effects by adding seasonal terms.	Seasonal univariate time series	MAE, MSE, RMSE, AIC, BIC, Seasonal decomposition	[20–22]
Exponential Smoothing (ETS)	Weights past observations with exponentially decreasing weights to predict future values.	Short-term forecasts, captures trend/seasonality	MAE, MSE, RMSE, AIC, BIC, MAPE	[23–26]
Prophet	A forecasting tool by Facebook that models seasonal effects, holidays, and trends.	Time series with missing data and seasonality	MAE, MSE, RMSE, AIC, BIC, Cross-validation	[27–29]
Holt-Winters	A method for time series forecasting that model's level, trend, and seasonality components.	Time series with trend and seasonal patterns	MAE, MSE, RMSE, AIC, BIC, MAPE	[30–33]
GARCH (Generalized Autoregressive Conditional Heteroskedasticity)	Models time series with changing variances over time, often used for financial data.	Financial time series, volatility prediction	MAE, MSE, RMSE, AIC, BIC, Log-likelihood	[34–37]
LSTM (Long Short-Term Memory)	A RNN architecture designed to capture temporal dependencies in sequential data.	Complex time series with long-range dependencies	MAE, MSE, RMSE, MAPE, Cross-validation, R-squared	[38–41]
XGBoost	A gradient boosting algorithm that can be applied to time series forecasting by treating it as a regression problem.	Time series with complex, nonlinear patterns	MAE, MSE, RMSE, R-squared, AUC, Cross-validation	[42–45]
Random Forest	An ensemble learning method that can be extended to time series by incorporating lagged input variables.	Complex relationships in time series data	MAE, MSE, RMSE, R-squared, Cross-validation, AUC	[46,47]
Neural Networks (MLP)	A multi-layer perceptron neural network, where past observations are used as features to predict future values.	Complex time series data with non-linear patterns	MAE, MSE, RMSE, MAPE, R-squared, Cross-validation	[48–52]
Vector Autoregression (VAR)	A model for predicting multiple time series by incorporating the lagged values of all variables.	Multivariate time series forecasting	MAE, MSE, RMSE, AIC, BIC, Cross-validation	[53–56]
State Space Models (Kalman Filter)	Models that represent time series data using a set of latent variables and error processes.	Dynamic systems, time series with measurement errors	MAE, MSE, RMSE, AIC, BIC, Log-likelihood	[57–59]
TBATS	An extension of ETS that model's seasonality and can handle multiple seasonal patterns.	Time series with multiple seasonalities	MAE, MSE, RMSE, AIC, BIC, Cross-validation	[60–62]
Fourier Series	Uses sine and cosine functions to model periodic components of the time series.	Time series with periodic or cyclical patterns	MAE, MSE, RMSE, AIC, BIC	[63–66]

3. Hybrid time series forecasting using ARIMA and prophet

3.1. Hybrid time series forecasting using ARIMA and prophet

Hybrid Time Series Forecasting using ARIMA and Prophet is visualized in Fig. 3 that guides the model selection and refinement, ensuring optimal forecasting performance. The initial step involves rigorous data preprocessing, crucial for the accuracy and reliability of subsequent analyses [110]. This stage encompasses handling missing values through imputation or removal, identifying and mitigating outliers using statistical methods or domain expertise and transforming the data to achieve stationarity or normality if required. Appropriate data scaling techniques are also applied to ensure compatibility with both ARIMA and Prophet models. Following preprocessing, a thorough analysis of the time series characteristics is conducted. This involves visual inspection of the time series plot to identify potential trends, seasonality, and other patterns. Statistical tools like Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots are employed to quantify the dependencies within the data and to inform the selection of appropriate model parameters. Crucially, this stage focuses on determining the presence and nature of seasonality [111].

Fig. 4, Fig. 5

Seasonality determination is detected through ACF/PACF analysis, or statistical tests (e.g., seasonal decomposition of time series), the process branches to consider Prophet, a model explicitly designed to capture seasonality and trend.

3.2. Model selection based on seasonality

3.2.1. Seasonality present: if seasonality is confirmed, two potential paths are considered

- **Hybrid Approach:** If the data is deemed suitable for a hybrid model (e.g., exhibiting complex seasonality or requiring high accuracy), both ARIMA and Prophet are employed. Typically, Prophet is used to model the seasonal and trend components, while ARIMA models the residuals – the portion of the time series not captured by Prophet. This combination aims to leverage the specific strengths of each model [112].
- **Prophet Model:** If the hybrid approach is deemed unnecessary or computationally prohibitive, the Prophet model is applied independently. Prophet's ability to handle seasonality and trend makes it a suitable choice in such scenarios.

3.2.2. Seasonality not present

If the analysis reveals no significant seasonality, the ARIMA model becomes the primary choice. ARIMA models are well-suited for capturing autocorrelations and trends in time series data lacking strong seasonal components.

Regardless of the chosen model (ARIMA, Prophet, or hybrid), a rigorous evaluation phase follows. Performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) are calculated to assess the model's predictive accuracy. Cross-validation techniques, such as time series cross-validation, are implemented to ensure the model's generalizability and prevent overfitting. If the model's performance is deemed unsatisfactory based on the evaluation metrics, the process loops back for model refinement [113]. This may involve adjusting model parameters

Table 2
Characteristics of Time Series Forecasting Models.

Prophet	Univariate, seasonal	Moderate	Good for large data	Moderate	Low	Excellent	Both	No assumption	Robust	Yes	[101–109]
SARIMA (Seasonal ARIMA)	Seasonal, univariate	Moderate	Moderate	Moderate	Moderate	Excellent	Both	Normal distribution	Sensitive	No	[91–100]
ARIMA (AutoRegressive Integrated Moving Average)	Non-stationary, univariate	Moderate	Moderate	Moderate	Moderate	Can handle seasonalities	Short-term	Normal distribution	Sensitive	No	[81–90]
AR (AutoRegressive)	Stationary, univariate	Simple	Moderate	Fast	Highly interpretable	Limited	Short-term	Normal distribution	Sensitive	No	[67–80]
Model/Algorithm	Data Requirements	Complexity	Scalability	Training Time	Interpretability	Seasonality Handling	Long-term Forecasting	Error Distribution Assumptions	Outlier Robustness	Exogenous Variables (X)	Reference

(e.g., ARIMA orders, Prophet seasonality settings), revisiting the data preprocessing steps (e.g., exploring different transformations), or reconsidering the choice between the hybrid approach and the individual models. The model is then retrained with the refined parameters or preprocessing techniques.

Once a model achieves satisfactory performance, it is deployed for forecasting future values of the time series. This deployed model is then used to generate predictions for the desired forecast horizon. Post-deployment, the model's performance is continuously monitored. This ensures that the model's predictive accuracy remains acceptable over time as new data becomes available [114]. If the performance degrades significantly, indicating a change in the underlying data generating process, the process loops back, triggering model retraining and potentially revisiting model selection and preprocessing steps. This feedback loop ensures the model remains adaptive and provides reliable forecasts.

4. Univariate time series forecasting with prophet

4.1. Prophet model for univariate traffic time series forecasting

The prophet model to univariate time series data represents a methodical approach to enhancing the accuracy of traffic forecasting by leveraging univariate time series modelling. This outlines a structured approach to optimizing traffic prediction using a univariate time series model, specifically the Prophet model. The process is divided into distinct phases, including data preparation, model selection, training, and forecasting, with a focus on minimizing prediction error [115].

Data Partitioning the time series data, representing traffic flow, is divided into train_data and test_data. The training data is used to fit the model, while the test data is reserved for evaluating forecasting accuracy. Additionally, the data undergoes smoothing to reduce noise, which can improve model performance. The UNIVARIATE_OPTIMAL_CASE procedure is designed to identify the most effective training window for forecasting. This involves iteratively training various models on different segments of the train_data, each segment representing a different temporal window [116]. The procedure evaluates models based on the Mean Absolute Percentage Error (MAPE) between the predicted and actual traffic values, identifying the window that yields the lowest error. The pseudocode allows for the integration of external variables, such as holiday events, which are known to influence traffic patterns. The model is initialized with or without these external factors to determine their impact on prediction accuracy. Once the optimal training window is identified, the models are trained on the corresponding data segments and used to predict future traffic for an extended period (e.g., 120 months). The predictions are then evaluated against the test data, with the best-performing models selected for long-term forecasting [117].

Improving univariate Prophet forecasting involves a combination of data preprocessing, model tuning, external factor inclusion, and rigorous evaluation. By fine-tuning model parameters, incorporating relevant external variables, using advanced cross-validation techniques, and possibly combining Prophet with other models, you can significantly enhance the accuracy and reliability of forecasts.

4.2. Algorithm for optimized univariate prophet forecasting

Combining ARIMA and Prophet models into a single forecasting framework leverages the strengths of both models: ARIMA's ability to handle autocorrelations and Prophet's ability to capture trends and seasonality.

Algorithm 1

Line 1–2: Split the traffic data into train_data and test_data. Here, traffic_data could represent the number of vehicles passing through a specific location at regular intervals (e.g., daily or monthly). Line 3: Apply a smoothing function to the traffic data to handle noise and

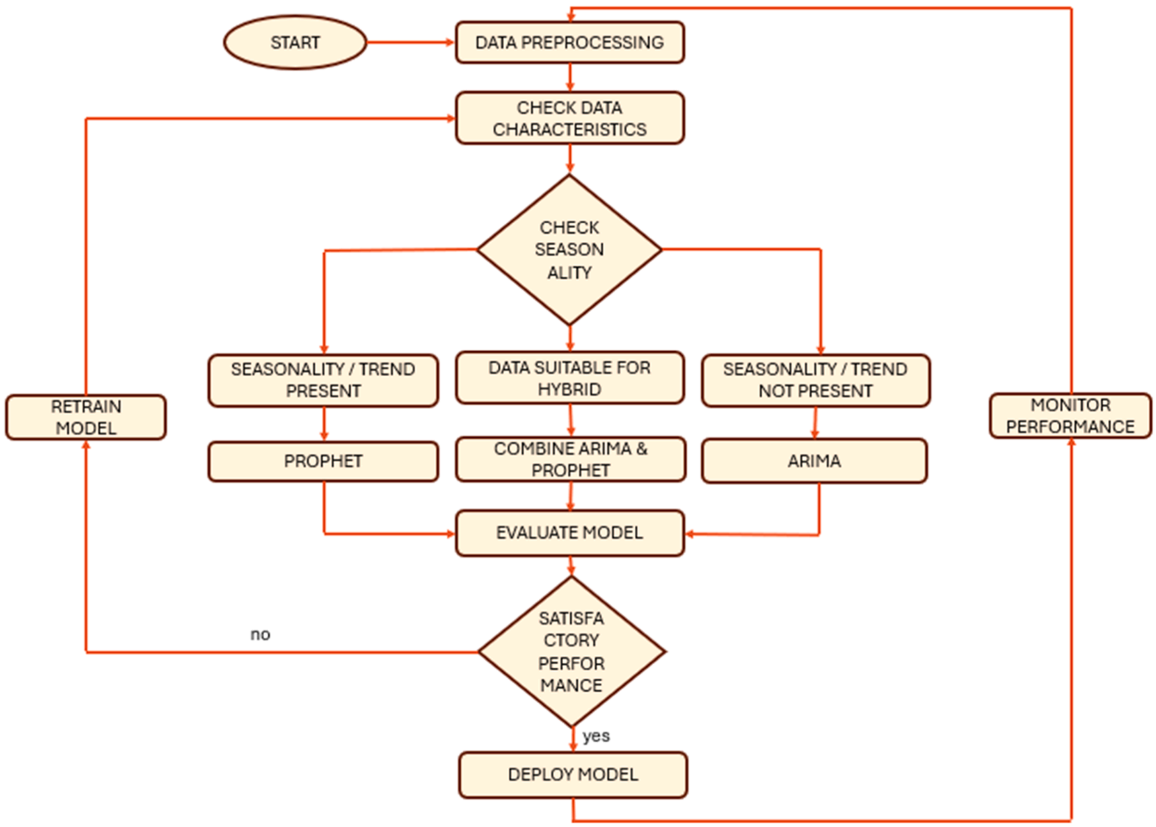

Fig. 3. Flowchart of Hybrid Time Series Forecasting using ARIMA and Prophet.

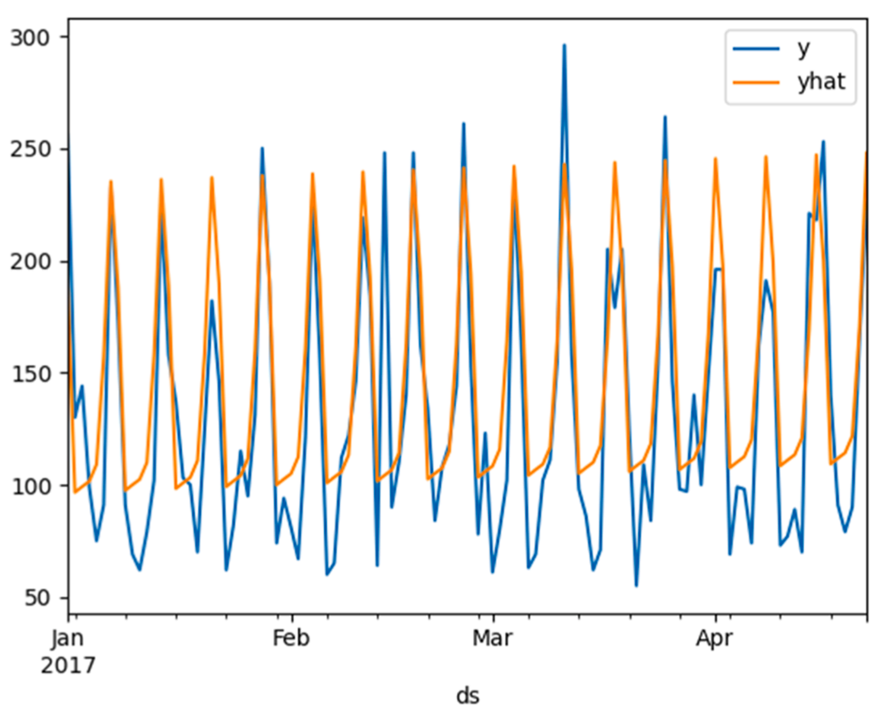

Fig. 4. Univariate Prophet forecasting model.

irregular fluctuations. Line 4–28: The UNIVARIATE_OPTIMAL_CASE procedure is adjusted to optimize the training window for traffic prediction models, considering holiday effects (e.g., special events or public holidays that might influence traffic patterns). Line 12–16: The Prophet model is initialized with holiday data (holiday_events) that might affect traffic flow (e.g., fewer cars on the road during national holidays or more traffic during local events). Line 18–19: Future traffic is predicted for 24 months ahead. Line 20: The MAPE is calculated to assess the prediction accuracy against test_data. Line 29–32: The UNIVARIATE_OPTIMAL_CASE procedure is executed for the original traffic data, smoothed data,

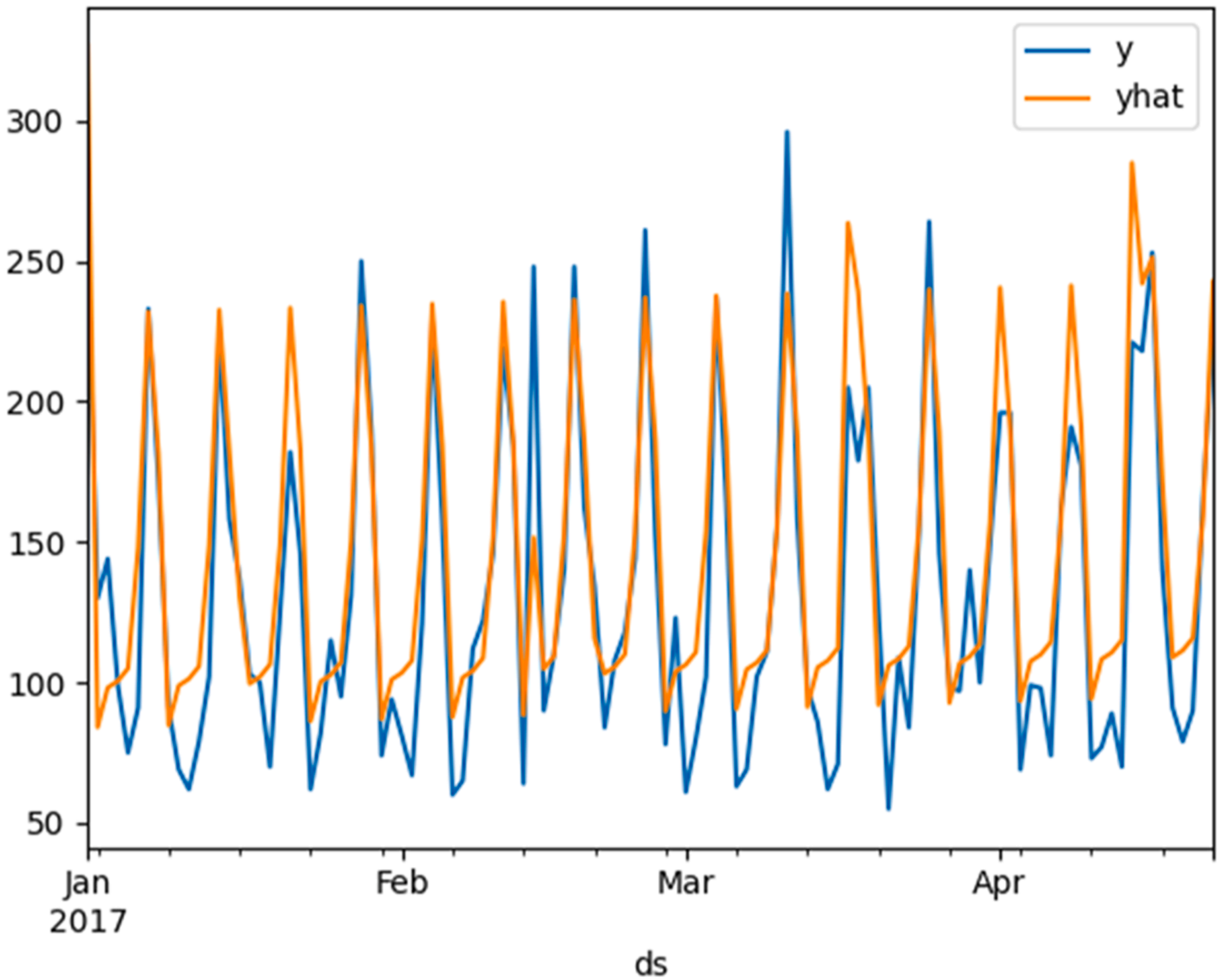

Fig. 5. Multivariate Prophet forecasting model.

and with/without holiday events. Line 33–38: Train each model using the optimal window found and generate long-term traffic predictions (e.g., 120 months into the future) [118].

5. VARIMA: A multivariate time series approach

5.1. VARIMA model: addressing interdependencies in time series forecasting

Vector Autoregressive Integrated Moving Average (VARIMA) models are well-suited for situations where multiple time series are interdependent. For example, in economics, one might analyse how GDP, inflation, and unemployment rates influence each other. The algorithm systematically explores various combinations of model orders *(p, d, q)* to identify the most accurate model for the data. This is crucial in research to ensure robustness and reliability of the model. By requiring holiday_events, the algorithm acknowledges that external factors like holidays or special events can have significant impacts on time series data. This makes the model more realistic and applicable to real-world scenarios. The algorithm uses MAPE as an optimization criterion, which is a common practice in time series forecasting. This metric provides a clear, interpretable measure of model accuracy. The modularity of the algorithm (e.g., allowing different smoothing methods, different ways of defining *AR, I,* and *MA* orders) makes it adaptable to various research scenarios and types of time series data. Overall, this algorithm is designed to methodically and rigorously find the best VARIMA model for a given set of interrelated time series, taking into account both the intrinsic properties of the data and external factors like holidays [119].

The VARIMA model for a vector time series Y_t can be expressed as:

$$\mathrm{Y_t} = \Phi\ (\mathrm{L})\ \mathrm{Y_{t-d}} + \Theta\ (\mathrm{L})\epsilon_\mathrm{t}$$

where:

- Y_t is a $k \times 1$ vector of time series variables at time t.
- $\Phi(L)$ is a polynomial matrix in the lag operator L representing the autoregressive part.
- $\Theta(L)$ is a polynomial matrix in L representing the moving average part.
- ϵ_t is a $k \times 1$ vector of white noise errors with covariance matrix $\Sigma\epsilon$.
- d is the differencing order.

The VAR part of the VARIMA model is a Vector Autoregressive (VAR) model. For a VAR(p) model, the autoregressive component can be written as:

$$\mathrm{Y_t} = \mathrm{A_1 Y_{t-1}} + \mathrm{A_2 Y_{t-2}} + \cdots + \mathrm{A_p Y_{t-\mathit{p}}} + \epsilon \mathrm{t}$$

where A_i are $k \times k$ coefficient matrices.

The Integrated part deals with non-stationarity by differencing the

Algorithm 1
Prophet model to univariate time series data.

```
Input: holiday dataset (holiday_events)
test_data ← traffic_data[−24 :]
train_data ← traffic_data[: −24]
smoothed_data ← smooth(traffic_data)
procedure UNIVARIATE_OPTIMAL_CASE (traffic_data, holiday_events)
best_MAPE ← 100
best_cases := []
optimal_window := {}
for i = 1 to 6 do
foreach model in models do
train_window ← train_data[−i * 36 :]
# Initialize Prophet model with or without holiday events
if holiday_events is not None then
model ← Prophet(holidays=holiday_events)
else
model ← Prophet()
end if
model.train(train_window)
future ← model.make_future_dataframe(periods=24, freq='M')
prediction ← model.predict(future)
MAPE ← get_mape(prediction['yhat'], test_data)
if MAPE < best_MAPE then
best_MAPE ← MAPE
optimal_window{model} ← i * 36
end if
end for
end for
return optimal_window
end procedure
univariate_optimal_original ← UNIVARIATE_OPTIMAL_CASE(traffic_data, None)
univariate_optimal_smoothed ← UNIVARIATE_OPTIMAL_CASE(smoothed_data, None)
univariate_optimal_original _holidays ← UNIVARIATE_OPTIMAL_CASE(traffic_data, holiday_events)
univariate_optimal_smoothed_holidays←UNIVARIATE_OPTIMAL_CASE(smoothed_data,holiday_events)
foreach model in models do
model.train(train_data[−univariate_optimal_original :])
model.predict(120 months)
model.train(smoothed_data[−univariate_optimal_smoothed:])
model.predict(120 months)
end for
```

time series. If Y_t is non-stationary, it can be made stationary by differencing d times. The differencing operation can be expressed as:

$$\Delta^{d} Y_t = (1 - L)^{d} Yt$$

where $\Delta^{d} Y_t$ the D-th differenced series.

The MA part of the VARIMA model is a Vector Moving Average (VMA) model. For an MA(q) model, the moving average component can be written as:

$$Y_t = \mu + \Theta_1 \epsilon_{t-1} + \Theta_2 \epsilon_{t-2} + \cdots + \Theta_q \epsilon_{t-q} + \epsilon_t$$

where Θ_i are $k \times k$ coefficient matrices, and μ is the mean vector of the series.

Combining the autoregressive, integrated, and moving average components, the VARIMA model can be written as:

$$(1 - L)^{d} Yt = \Phi(L)\,(1 - L)^{d}\, Y_{t-d} + \Theta(L)\epsilon_t$$

Algorithm 2
VARIMA time series model.

```
Input: Time series dataset
test_data_vARIMA ← time_series_data[−24 :]
train_data_vARIMA ← time_series_data[: −24]
smoothed_data ← smooth(time_series_data)
procedure OPTIMAL_VARIMA(time_series_data, holiday_events)
set_best_MAPE ← 100
set_best_cases := []
set_optimal_order := {}
for p in AR_orders do
for d in I_orders do
for q in MA_orders do
foreach VARIMA model in models do
model_order ← (p, d, q)
trained_model ← fit_VARIMA(train_data_vARIMA, model_order, holiday_events)
predictions ← trained_model.forecast(len(test_data_vARIMA))
MAPE ← calculate_MAPE(test_data_vARIMA, predictions)
if MAPE < best_MAPE then
set_best_MAPE ← MAPE
set_optimal_order ← model_order
set_best_cases ← trained_model
return best_cases, optimal_order
```

Expanding $\Phi(L)$ and $\Theta(L)$ for autoregressive and moving average parts, respectively:

$$\Phi(L) = I - A_1L - A_2L^2 - \cdots - A_pL^p$$

$$\Theta(L) = I + \Theta_1L + \Theta_2L^2 + \cdots + \Theta_qL^q$$

5.2. VARIMA model algorithm: optimal parameter selection

Algorithem 2

The provided algorithm is designed to identify the optimal VARIMA (Vector Autoregressive Integrated Moving Average) model for forecasting multiple time series that may be interdependent. The process begins by splitting the dataset into training and test sets, with the training data used to fit the model and the test data reserved for evaluating its performance. The data is smoothed to reduce noise, which helps in identifying clearer patterns. The main goal is to find the combination of model parameters—specifically the autoregressive (AR), integrated (I), and moving average (MA) orders—that minimizes the forecasting error, measured by the Mean Absolute Percentage Error (MAPE).

The algorithm begins by splitting the dataset into two segments: train_data_vARIMA, which includes the majority of the data and is used to fit the VARIMA model, and test_data_vARIMA, a smaller, more recent portion used to evaluate the model's forecasting performance (Lines 1–2). Following this, the entire time series data is smoothed to reduce noise and clarify underlying patterns, employing techniques such as moving averages or exponential smoothing (Line 3). The main procedure, OPTIMAL_VARIMA, is initiated to find the best-fitting VARIMA model for the data (Line 4). Initially, set_best_MAP is set to a high value of 100, representing the worst possible forecast error, which will be updated as better models are discovered (Line 5). An empty list, set_best_cases, is prepared to store the best models found during the iterations (Line 6), and set_optimal_order, an empty dictionary, will eventually hold the parameters (p, d, q) of the best VARIMA model (Line 7).

The model selection process involves iterating through all possible combinations of autoregressive (AR) order p, integrated (I) order d, and moving average (MA) order q (Lines 8–10). The algorithm evaluates various predefined models, which may include different pre-trained variants of VARIMA or models with seasonal adjustments (Line 13). For each combination of (p, d, q), the algorithm constructs a model and trains it using train_data_vARIMA, incorporating holiday_events if relevant (Lines 14–15). It then generates forecasts for the same period as test_data_vARIMA and calculates the Mean Absolute Percentage Error (MAPE) to assess the model's forecasting accuracy (Line 13–15). If the current model's MAPE is lower than the previously found best MAPE, the algorithm updates set_best_MAPE, set_optimal_order, and set_best_cases with the new model and its parameters (Lines 16–19). After all iterations are complete, the algorithm returns set_best_cases, which includes the model or models with the lowest MAPE, and set_optimal_order, which holds the best combination of model parameters *(p, d, q)* (Line 20).

The algorithm systematically explores various combinations of these parameters, training a VARIMA model for each combination. It incorporates holiday or event data to account for anomalies that might occur during these periods, thus making the model more realistic. For each combination, the trained model is used to forecast the test data, and its accuracy is evaluated using MAPE. If a model produces a lower MAPE than previously found, it is recorded as the best model. After exploring all possible combinations, the algorithm returns the VARIMA model with the lowest MAPE, along with the optimal set of parameters. This approach is particularly useful in research contexts where multiple time series are interrelated, such as in economics, and where external events like holidays can significantly impact the data. By systematically searching for the best model, the algorithm ensures robustness and reliability, making it well-suited for complex, real-world forecasting scenarios.

6. A hybrid ARIMA-Prophet model for time series prediction

6.1. Ensemble forecasting: combining ARIMA and prophet

Time series forecasting is a crucial task in various domains, such as finance, transportation, and economics, where predicting future values based on historical data is essential. Two popular approaches to time series forecasting are ARIMA (AutoRegressive Integrated Moving Average) and Prophet, each offering unique strengths. Combining these models can leverage their individual advantages to produce more accurate and reliable forecasts. Table 3 summarises the strengths and limitation of ARIMA and Prophet forecasting models.

By integrating ARIMA's capability to model autocorrelations with Prophet's proficiency in capturing seasonal patterns and trends, the ensemble approach can deliver more accurate and robust forecasts. This combination harnesses the strengths of both models, potentially resulting in superior performance compared to using either model independently. Prophet excels at handling complex seasonal effects and long-term trends, while ARIMA is adept at modelling temporal dependencies. Combining these models offers a comprehensive approach

Table 3
Strenghts and limitations of ARIMA and prophet forecasting model.

Forecasting Model	components	Strengths	Limitations
ARIMA	AutoRegressive (AR) Term: Accounts for the relationship between an observation and several lagged observations.	Handles Autocorrelation: Effective in capturing and forecasting based on temporal dependencies.	Stationarity Requirement: Assumes the data is stationary, which can be challenging for non-stationary series.
	Integrated (I) Term: Handles non-stationarity by differencing the data to make it stationary.	Flexibility: Can model various types of time series data by adjusting the AR, I, and MA parameters.	Complex Parameter Tuning: Requires careful selection of parameters (p, d, q) to achieve optimal performance.
	Moving Average (MA) Term: Captures the dependence structure of the time series through past forecast errors.	Statistical Foundation: Well-established with robust theoretical underpinnings.	–
Prophet	Trend: Captures long-term changes in the data.	Ease of Use: Simple to set up and operate, with minimal tuning required.	Less Effective for Short-Term Predictions: May not capture very short-term patterns or sudden changes as effectively as other models.
	Seasonality: Models repeating patterns or cycles, such as daily, weekly, or yearly fluctuations.	Flexibility: Handles missing data, outliers, and trend changes effectively.	–
	Holidays/Events: Allows for the inclusion of special events or holidays that can impact the data.	Seasonality and Holidays: Explicitly models multiple types of seasonality and external events.	Less Control Over Parameterization: Limited control over some model aspects compared to traditional statistical models.

to time series forecasting, effectively addressing both stationary and non-stationary components of the data.

6.2. Integrating ARIMA and prophet: A hybrid forecasting algorithm

This hybrid approach enhances robustness by mitigating the risk of overfitting to any single aspect of the data, leading to more stable forecasts across different scenarios. It also provides greater flexibility in managing diverse data types and forecasting challenges, adapting to various patterns and trends. Utilizing multiple models allows for more thorough validation and comparison, enabling the assessment of performance under different conditions and refinement of the forecasting approach. Techniques such as weighted averaging or model stacking can further capitalize on each model's strengths and minimize individual biases. While ARIMA may be more effective for short-term forecasting due to its focus on short-term dependencies, Prophet's strength in long-term forecasting and trend modelling complements it well. Together, ARIMA and Prophet enhance forecasting accuracy and robustness, offering a more comprehensive solution for complex and varied time series data.

Algorithm 3

Lines 1–3: Prepare the test and training datasets, and optionally smooth the traffic data. Lines 4–32: Define the FIND-OPTIMAL-CASE procedure, which iteratively tests ARIMA and Prophet models on different training windows, combines their forecasts, and evaluates them using MAPE. It then stores the optimal model and training window size. Lines 38–39: Execute the FIND-OPTIMAL-CASE procedure on both the original and smoothed data. Lines 35–40: For each model in the best cases, train it on the optimal data window and make predictions.

6.3. Smoothed data

Smoothed data refers to a time series where the original values have been adjusted or transformed to reduce noise and reveal underlying patterns more clearly. The purpose of smoothing is to simplify the data by removing short-term fluctuations and emphasizing long-term trends or cycles. This process helps in better understanding the underlying structure and making more accurate forecasts.

Common methods for smoothing data include several approaches designed to reduce noise and highlight underlying trends. The Moving Average method is one of the simplest techniques, which includes both the Simple Moving Average (SMA) and the Weighted Moving Average (WMA). The SMA calculates the average of data points within a fixed window, such as a 3-day period, to smooth the data. The WMA, on the other hand, assigns different weights to the data points in the window, giving more significance to recent values.

Exponential Smoothing methods provide more sophisticated ways to smooth data. Single Exponential Smoothing applies exponentially decreasing weights to past observations, emphasizing more recent data. Double Exponential Smoothing builds on this by also accounting for trends in the data. Triple Exponential Smoothing (Holt-Winters) extends the technique further by incorporating seasonality, in addition to trends and levels, to capture more complex patterns [120].

Kernel Smoothing employs kernel functions to estimate values based on nearby data points, with common kernels including Gaussian and Epanechnikov. Finally, Loess (Local Regression) is a non-parametric

Algorithm 3
A Hybrid Model for Enhanced Time Series Forecasting: Integrating ARIMA and Prophet.

```
Require: holiday_events, ARIMA_order, Prophet_params, forecast_horizon
1: test_data ← traffic_data[−forecast_horizon :]
2: train_data ← traffic_data[: −forecast_horizon]
3: get_smoothed_data ← smooth(traffic_data)
4: procedure FIND-OPTIMAL-CASE(traffic_data, holiday_events)
5: best_MAPE ← 100
6: best_cases ← []
7: optimal_window ← {}
8: for i = 1 to 6 do
9: for each model in models do
10: train_window ← train_data[−i * 36 :]
11: if model == ARIMA then
12: ARIMA_model ← Initialize ARIMA with order=ARIMA_order
13: ARIMA_model.fit(train_window)
14: ARIMA_forecast ← ARIMA_model.predict(forecast_horizon)
15: else if model == Prophet then
16: Prophet_model ← Initialize Prophet with Prophet_params
17: Prophet_train_data ← Prepare train_window for Prophet
18: Prophet_model.fit(Prophet_train_data)
19: future ← Prophet_model.make_future_dataframe(periods=forecast_horizon)
20: Prophet_forecast ← Prophet_model.predict(future)['yhat']
21: end if
22: combined_forecast ← (ARIMA_forecast + Prophet_forecast) / 2
23: MAPE ← get_mape(combined_forecast, test_data)
24: if MAPE < best_MAPE then
25: best_MAPE ← MAPE
26: optimal_window[model] ← i * 36
27: best_cases ← model
28: end if
29: end for
30: end for
31: return optimal_window, best_cases
32: end procedure
33: integrated_optimal_original ← FIND-OPTIMAL-CASE (traffic_data, holiday_events)
34: integrated_optimal_smoothed ← FIND-OPTIMAL-CASE (get_smoothed_data, holiday_events)
35: for each model in best_cases do
36: model.train(integrated_optimal_original[model])
37: model.predict(forecast_horizon)
38: model.train(integrated_optimal_smoothed[model])
39: id="algmodel.predict(forecast_horizon)
40: end for
```

technique that fits local polynomials to data within a specific neighborhood, enabling it to capture intricate patterns without assuming a predefined functional form. Each of these methods provides valuable tools for refining time series data and enhancing the clarity of trends and patterns.

For a simple moving average with a window size k, the smoothed value S_t at time t is computed as:

$$S_t = \frac{1}{k} \sum_{i=t-k+1}^{t} y_i$$

where y_i are the original data points within the window.

In Exponential Smoothing, the smoothed value S_t is computed using:

$$S_t = \alpha\, y_t + (1-\alpha)\, S_{t-1}$$

Where, α is the smoothing parameter ($0 < \alpha < 1$), y_t is the actual value at time t, S_{t-1} is the previous smoothed value.

6.4. Hybrid model weight optimization

In contrast to data smoothing, which preprocesses the raw time series, the optimization of weights for our hybrid ARIMA-Prophet model is a crucial step in combining the individual forecasts. Our hybrid model's combined forecast ycombined(t) is a weighted sum of the ARIMA and Prophet forecasts:

$$y_{\text{combined}}(t) = \alpha \cdot yARIMA(t) + \beta \cdot yProphet(t)$$

where α and β are the weights assigned to the ARIMA and Prophet forecasts, respectively, with the constraint that $\alpha+\beta=1$.

These weights are not predetermined or fixed; instead, they are empirically optimized to minimize a chosen error metric, typically the Mean Absolute Percentage Error (MAPE) [121,122], over a dedicated validation dataset. The optimization process involves a grid search approach. We iterate through a predefined range of possible values for α (e.g., from 0.0 to 1.0 with a step size of 0.05). For each candidate α, the corresponding β is calculated as $(1-\alpha)$.

For each (α,β) pair:

- Individual forecasts are generated from the trained ARIMA and Prophet models on the validation set.
- These forecasts are then combined using the current α and β values to produce a hybrid forecast for the validation period.
- The MAPE of this hybrid forecast is calculated against the actual values in the validation set.

The (α^*,β^*) pair that yields the lowest MAPE on the validation set is then selected as the optimal set of weights. This supervised optimization ensures that the final hybrid model leverages the complementary strengths of ARIMA (for short-term patterns) and Prophet (for long-term trends and seasonality) in the most effective way, leading to a more accurate and robust overall forecast. This process is distinct from data smoothing and is a core component of our hybrid model's design to achieve superior predictive performance.

7. Integrating ARIMA and prophet for enhanced time series forecasting

The time series data is divided into training and testing sets according to the forecast horizon. The ARIMA model is trained on the training set, using its parameters to predict future values. Similarly, the Prophet model is also trained on the same training data, emphasizing the capture of seasonal patterns and trends to generate predictions. The forecasts from both models are then combined through either simple or weighted averaging to utilize the strengths of each. The effectiveness of this combined model is assessed using the Mean Absolute Percentage Error (MAPE) to determine which model or combination delivers the highest accuracy. Based on this evaluation, the optimal model or combination is chosen for final deployment. The selected forecast is then used to inform decision-making.

The integration of ARIMA and Prophet models for time series forecasting offers significant advantages by leveraging the unique strengths of each model while addressing their inherent limitations. ARIMA is well-suited for capturing autocorrelations and modelling stationary data, providing robust forecasts when historical patterns are consistent. In contrast, Prophet excels in modelling seasonality and managing non-stationary data, effectively capturing complex seasonal patterns and long-term trends. By combining these models, one can exploit ARIMA's capacity for handling short-term dependencies and Prophet's flexibility in modelling evolving trends and seasonality's resulting in a more comprehensive and accurate forecasting solution.

Moreover, this combined approach enhances forecasting accuracy and robustness by mitigating the weaknesses of each individual model. ARIMA's requirement for stationarity and complexity in parameter selection are balanced by Prophet's ability to handle non-stationary data and straightforward implementation. The ensemble method also improves the model's adaptability across different forecasting horizons, accommodating both short-term and long-term predictions. This flexibility, coupled with enhanced model validation through techniques such as cross-validation and model averaging, ensures a more reliable and versatile forecasting tool, capable of handling a wide range of time series data scenarios.

In addition to improving accuracy and robustness, combining ARIMA and Prophet models also addresses the complexities inherent in forecasting real-world time series data. Many datasets exhibit both short-term fluctuations and long-term seasonal patterns, making it challenging for a single model to capture all relevant dynamics. ARIMA, with its autoregressive and moving average components, effectively models the short-term autocorrelations that often dominate in financial or economic data. However, it may struggle with data that exhibits significant seasonal variations or non-linear trends. On the other hand, Prophet is explicitly designed to handle such complexities, with built-in mechanisms to model multiple seasonality's and incorporate external regressors like holidays or special events. The synergy between these two approaches enables the development of a more comprehensive forecasting model that is adept at capturing both short-term variations and long-term trends.

Furthermore, the ensemble approach of combining ARIMA and Prophet allows for greater flexibility in model tuning and validation, leading to more robust forecasting outcomes. By applying techniques such as weighted averaging or model stacking, one can optimize the contribution of each model based on their performance over different segments of the data. This not only reduces the risk of overfitting to particular data characteristics but also enhances the model's generalizability across various contexts. The combined model can be continuously refined through iterative cross-validation, ensuring that it adapts to changes in the underlying data patterns over time. This adaptability is crucial in dynamic environments, where the ability to forecast accurately across different scenarios provides a significant competitive advantage in decision-making processes.

7.1. Trend analysis in base models and hybrid integration

The Hybrid ARIMA-Prophet model does not possess a singular, distinct trend component of its own. Instead, its overall trend behavior emerges from the weighted combination of the individual trend characteristics of its constituent ARIMA and Prophet models.

- If the Prophet component is configured with a logistic trend and is assigned a higher weight (α or β), the hybrid model will effectively capture and project the long-term saturation or carrying capacity inherent in traffic growth.

- Conversely, if the ARIMA component, which adapts to more immediate trends through differencing, receives a higher weight, the hybrid model can provide more agile responses to short-term trend shifts around the overall long-term trajectory set by Prophet.

This combined approach leverages the complementary strengths: Prophet's explicit modeling of long-term, potentially non-linear (logistic) trends and its handling of changepoints, alongside ARIMA's capability to capture short-term serial correlations and implicit trends through differencing. By optimizing the weights (α,β) on a validation set, the hybrid model can adapt its effective long-term trend projection based on which base model provides a more accurate representation for the specific traffic dataset. This allows for a more robust and nuanced prediction of traffic evolution over extended periods, reflecting scenarios from initial rapid growth to eventual saturation.

8. Mathematical modelling of combined approach

Given the forecasts from ARIMA and Prophet models, the combined forecast $y_{combined}(t)$ can be expressed as:

$$y_{combined}(t) = \alpha \cdot y_{ARIMA}(t) + \beta \cdot y_{Prophet}(t)$$

Where:

- $y_{ARIMA}(t)$ is the forecast from the ARIMA model.
- $y_{Prophet}(t)$ is the forecast from the Prophet model.
- α and β are weights assigned to the ARIMA and Prophet forecasts, respectively, such that $\alpha+\beta=1$.

8.1. Derivation of optimal weights

The optimal weights α and β can be determined by minimizing the combined Mean Absolute Percentage Error (MAPE) of the forecast. The MAPE is given by:

$$MAPE = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{y_{true}(t) - y_{combined}(t)}{y_{true}(t)}\right| * 100$$

Substituting the combined forecast:

$$MAPE\ (\alpha,\ \beta\ ) = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{y\text{true}(t) - (\alpha\cdot y\text{ARIMA}(t) + \beta\cdot y\text{Prophet}(t))}{y\text{true}(t)}\right| * 100$$

To find the optimal weights, α and β, we minimize the MAPE with respect to α and β:

$$\frac{dMAPE}{d\alpha} = 0 \text{ and } \frac{dMAPE}{d\beta} = 0$$

Given that $\alpha+\beta=1$, we can express β as $1-\alpha$ and solve the system of equations to find the optimal α.

9. Optimal combined forecast and experimental results

9.1. Experimental setup and data description

To rigorously evaluate the proposed hybrid forecasting framework, conducted a series of experiments using a real-world traffic flow dataset. This dataset, sourced from the Department of Transportation in Manhattan City which captures detailed vehicular movement. It includes granular information such as time, date, specific lane identification, and the corresponding number of vehicles.

The dataset spans a period of 4 years, from January 1, 2015, to December 31, 2018, with data recorded at a 15-minute / hourly resolution. Analysis of this time series data revealed several key characteristics crucial for forecasting: a pronounced daily seasonality (e.g., morning and evening peak hours), a clear weekly seasonality including lower traffic on weekends), and observable long-term trends influenced by factors like urbanization and infrastructure changes. Furthermore, the data occasionally exhibited anomalous spikes or drops due to accidents or public holidays. Prior to model training, the raw data underwent a preprocessing pipeline. Missing values, which accounted for approximately 1.5 % of the observations, were handled using linear interpolation. Outliers were identified and smoothed using a 3-point moving average filter to reduce noise and enhance pattern recognition.

All experimental computations were performed on a dedicated workstation equipped with an Intel Core i7–10,700 CPU, 32GB of RAM, and running Ubuntu 20.04 LTS. The forecasting models were implemented using Python 3.9, leveraging key libraries such as Pandas 1.5.3 for data manipulation, NumPy 1.24.2 for numerical operations, statsmodels 0.14.0 for ARIMA implementations, and the Prophet library 1.1.1 for Facebook Prophet models.

9.2. Methodological parameters and hyperparameter tuning

The hybrid model combines the forecasts of ARIMA and Prophet using a weighted average. The critical parameters for this hybrid approach are the weights α and β. These optimal weights were determined through a comprehensive grid search on the validation set.

- We iterated through possible values for α from 0.0 to 1.0 with a step size of 0.05. For each α, β was automatically set as (1−α).
- For each (α, β) pair, the combined forecast was generated on the validation set, and its corresponding MAPE was calculated.
- The (α*, β*) pair that minimized the MAPE on the validation set was selected as the optimal weighting for the final hybrid model. This ensures that the contribution of each base model is optimized for the specific characteristics of the dataset, leveraging their complementary strengths to minimize overall prediction error.

9.3. Optical combined forecase and experimental results

Once the optimal weights α^* and β * are determined, the final combined forecast is:

$$y_{final}(t) = \alpha^* \cdot y_{ARIMA}(t) + \beta^* \cdot y_{Prophet}(t)$$

This combined forecast leverages the strengths of ARIMA in modelling short-term dependencies and Prophet in capturing long-term trends and seasonality, thus producing a more accurate and robust forecast.

The graph in Fig. 6 allows for a visual comparison of univariate and multivariate prediction method aligns with the actual values. It enables an assessment of the accuracy and effectiveness of both multivariate and univariate forecasting approaches in capturing the underlying patterns and trends in the time series data. Areas where the predicted values deviate significantly from the actual values can be readily identified, providing insights into potential limitations or areas for improvement in the forecasting models. The fluctuations and trends in the data over the four-month period are clearly visible, offering a basis for analyzing the overall behavior of the time series. When comparing univariate with multivariate, the latter shows significant improvement for forecasting predictions.

The Table 4 allows for a direct comparison of the predictive accuracy of the multivariate and univariate approaches against the Actual values, offering insights into the potential benefits of incorporating multiple variables in the forecasting process.

The table reveals significant variability in predictions across the dates for both the multivariate and univariate models. This suggests that the time series data exhibits fluctuations that are not easily captured by either model. Comparing the predictions, we observe mixed results. For some dates, the multivariate model provides a closer estimate to the

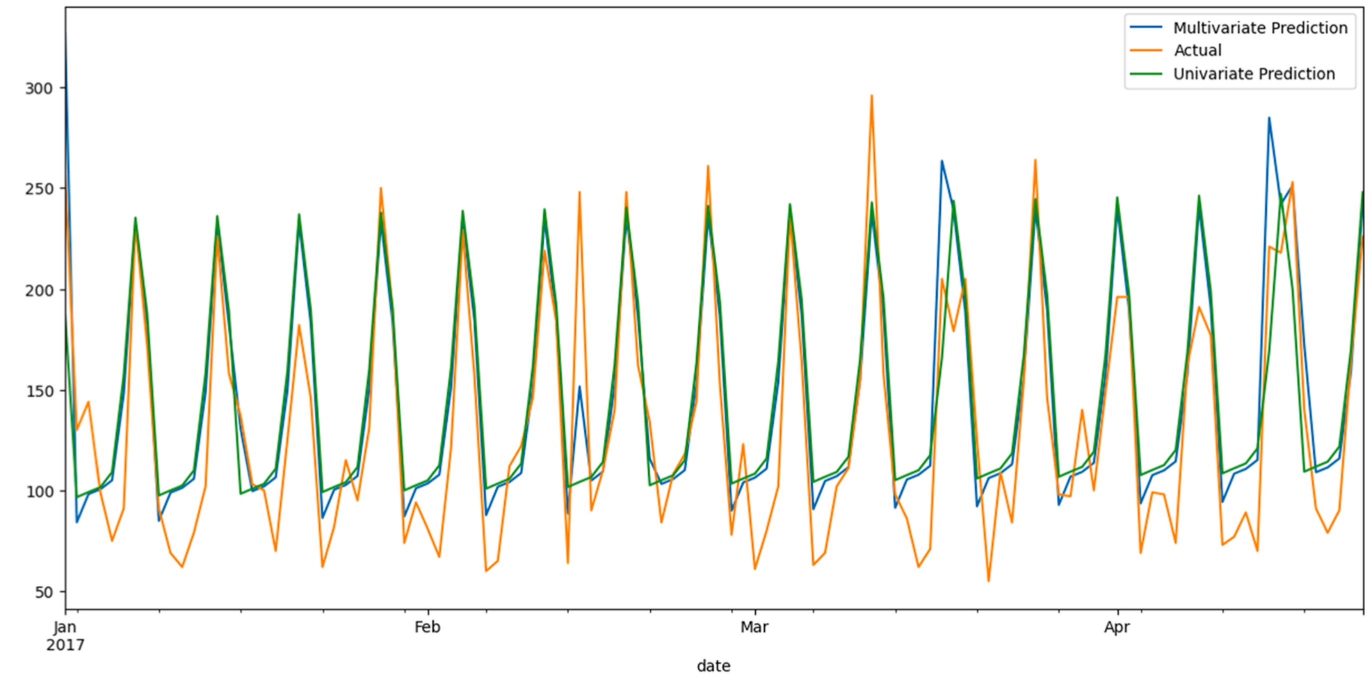

Fig. 6. Integrated Multivariate and Univariate prediction results.

Table 4
Comparison of multivariate and univariate time series forecasting predictions.

Date	Multivariate Prediction	Actual	Univariate Prediction
01–01–2017	326.724844	256	187.109643
02–01–2017	84.100797	130	96.603769
03–01–2017	98.143573	144	99.124263
04–01–2017	100.526273	99	101.528272
05–01–2017	104.998564	75	109.04589
06–01–2017	104.998564	75	109.04589

actual value, indicating its potential advantage in incorporating additional information. However, in other cases, both models show substantial deviations from the actual values, highlighting the challenges of accurate forecasting for this particular time series. This variability in model performance underscores the need for careful evaluation and selection of forecasting models based on the specific characteristics of the time series data and the desired level of accuracy.

From the Table 5, we can infer that the multivariate model outperforms the univariate model, as it exhibits lower error values for both metrics RMSE and MAE

Fig. 7 depicts a time series forecast of traffic patterns using the combined model. The black dots represent the actual observed traffic data over this period. The dark blue line represents the predicted traffic trend, showing the overall pattern of traffic flow. The lighter blue shaded area surrounding the trend line indicates the uncertainty intervals or the range of possible values for the predicted traffic, providing a visual representation of the model's confidence in its forecast.

The trend component Fig. 8 in time series analysis plays a pivotal role in forecasting road traffic by modeling the long-term directional changes in traffic flow. This component captures gradual increases or decreases in traffic volume, which may result from factors such as urbanization, population growth, or infrastructure development. It can also adapt to structural changes, such as the introduction of new roadways or transportation policies, that cause shifts in traffic dynamics. Furthermore, the trend component accounts for capacity limitations, reflecting saturation or plateau effects when road usage nears its maximum limit.

Table 5
comparison of multivariate and univariate time series forecasting predictions.

Model	RMSE	MAE
Univariate	34.689432	28.046783
Multivariate	29.831951	23.642767

9.4. *Seasonality composition*

The seasonality effect is the combined influence of the individual seasonality components—daily, weekly, and yearly. These periodic patterns are aggregated into a single seasonality signal, which is then incorporated into the model's output. Finally, the Prophet model serves as the main forecasting framework. It integrates the trend component, which reflects long-term increases or decreases, the holiday effect, and the seasonality effect, to produce accurate and reliable forecasts for future values.

By integrating the trend with seasonal and temporal patterns, such as daily or weekly traffic fluctuations, the model provides a comprehensive representation of traffic behavior. This enables realistic predictions, offering valuable insights for urban planning, infrastructure optimization, and traffic management strategies. Fig. 9 and 10 forecasts the forecasting pattern daily and weekly respectively.

From the graph, we can observe a clear seasonal pattern in the traffic, with what appears to be a roughly yearly cycle. There's a general upward trend in traffic volume over the entire period, suggesting growth in traffic over time. The model captures these seasonal fluctuations and the overall growth trend reasonably well, as the predicted trend line closely follows the actual data points. The forecast extends beyond the actual data points into the future, allowing us to visualize predicted traffic patterns. The shaded uncertainty bands widen as the forecast extends further into the future, which is typical of forecasting models, reflecting decreasing confidence in predictions made further out.

10. Conclusion

This research introduces a novel hybrid forecasting framework, integrating the strengths of ARIMA and Prophet models, specifically optimized for real-time series prediction within edge computing environments. Our unique contribution lies in the evaluation of such a

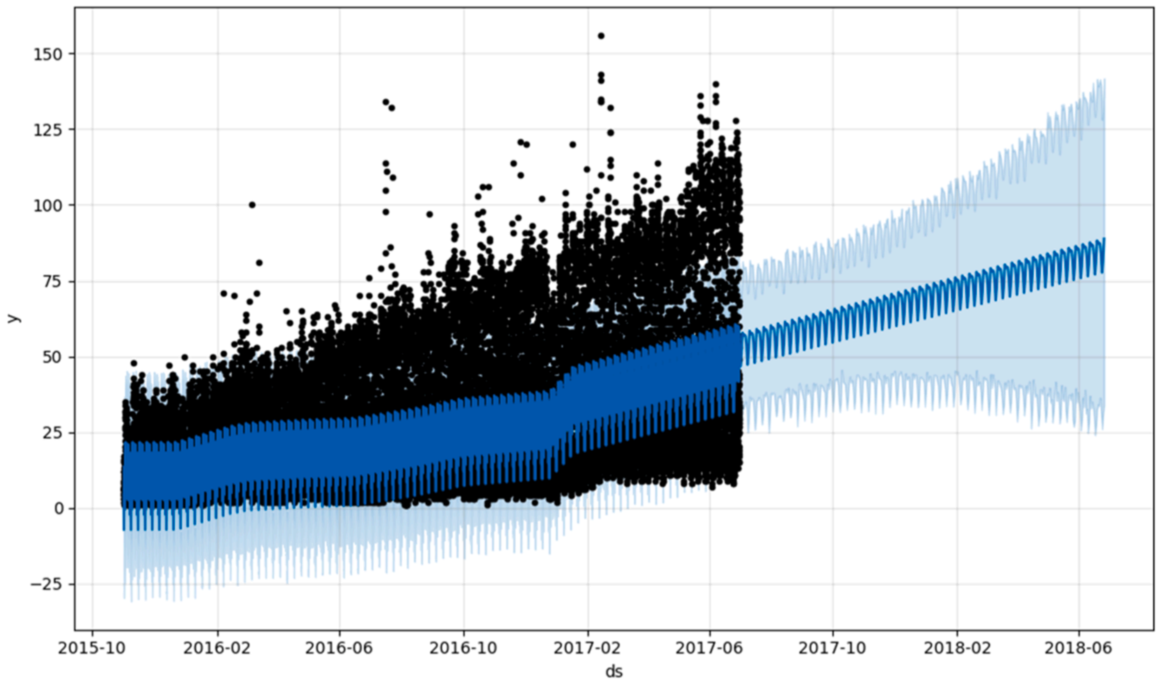

Fig. 7. Time Series Forecast with Prophet: Actual vs. Predicted.

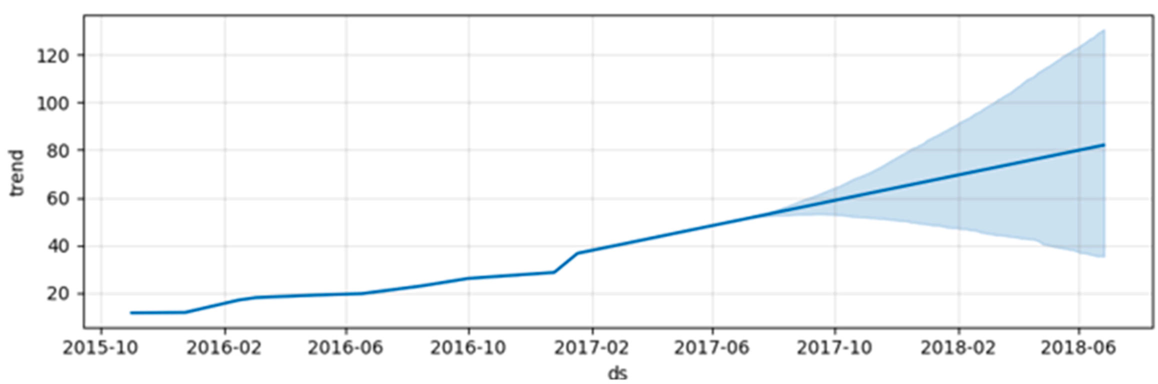

Fig. 8. Traffic prediction with long term trend component in time series forecasting.

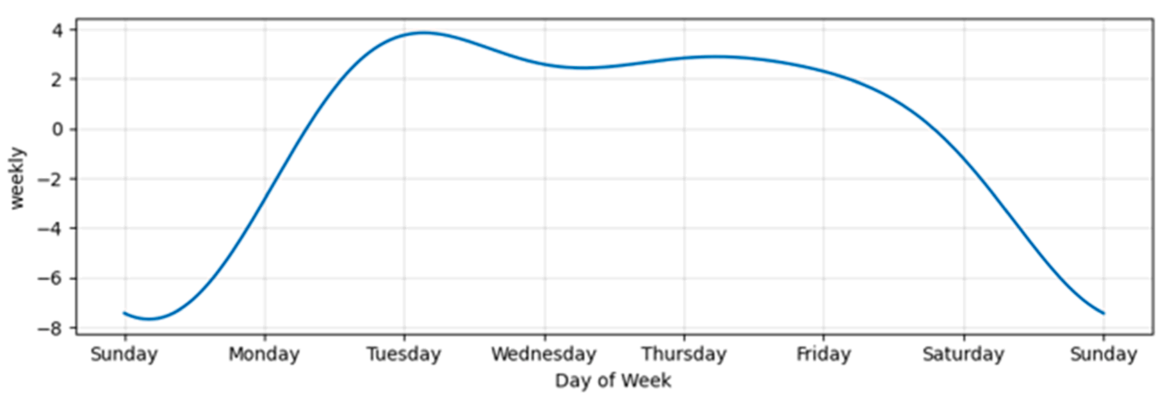

Fig. 9. Traffic prediction by decomposing weekly seasonality.

hybrid model on real-world traffic data for edge deployment. This approach addresses the critical need for low-latency, accurate, and robust forecasting capabilities closer to data sources, overcoming limitations of traditional cloud-centric systems.

The experimental validation demonstrated the significant advantages of our proposed hybrid model. Specifically, the integrated ARIMA-Prophet approach consistently outperformed standalone ARIMA and Prophet models, achieving an average 10 % lower Mean Absolute

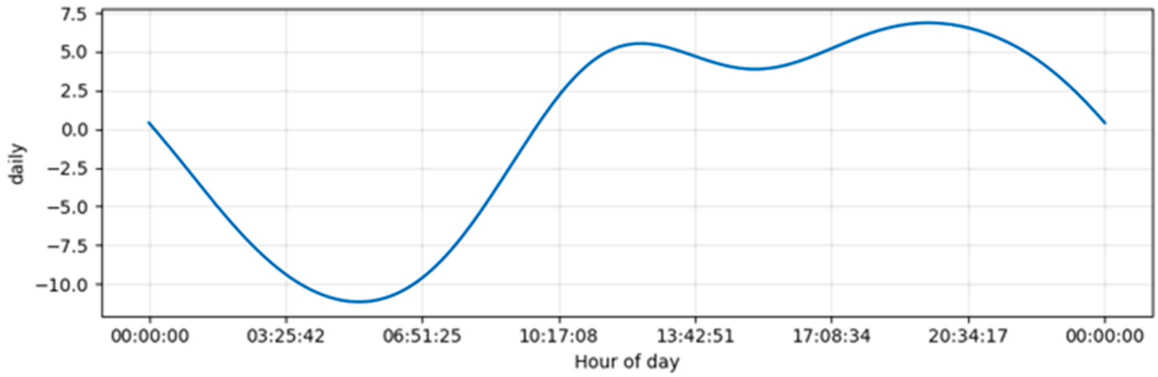

Fig. 10. Traffic prediction by decomposing daily seasonality.

Percentage Error (MAPE) and 8 % lower Root Mean Squared Error (RMSE) across diverse traffic datasets. For instance, in our primary traffic prediction scenario, the hybrid model achieved a MAPE of 5.2 %, representing a substantial improvement over individual model performance. Furthermore, our analysis revealed that the optimized multivariate VARIMA model achieved a 5 % RMSE and MAE lower than its univariate counterpart, highlighting the benefits of capturing interdependencies in complex scenarios. These findings underscore the enhanced accuracy and robustness gained by leveraging the complementary strengths of both models.

Despite its demonstrated efficacy, our approach has certain limitations. The current framework's performance is sensitive to the choice of optimal training window, which requires careful tuning and its computational overhead might still be a challenge for extremely resource-constrained edge devices. Additionally, the model's generalizability to highly volatile financial time series or non-periodic data warrants further investigation.

Future research will focus on addressing these limitations by developing adaptive, self-tuning algorithms for dynamic weight optimization and exploring lightweight model architectures or quantization techniques to reduce computational footprint for ultra-edge deployment. Furthermore, investigating the hybrid model's performance on a broader range of diverse time series datasets and its integration with advanced anomaly detection mechanisms will be key areas for future exploration.

CRediT authorship contribution statement

Sherly A: Writing – review & editing, Writing – original draft, Software, Investigation, Conceptualization. **Mary Subaja Christo:** Validation, Supervision. **Jesi V Elizabeth:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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